MTH357

Timed Online Assignment – January Semester 2020

Applied Regression Analysis I

Thursday, 14 May 2020 **4:00 pm – 6:30 pm**

Time allowed: 2.5 hours

INSTRUCTIONS TO STUDENTS:

1. This examination Timed Online Assignment (TOA) comprises **NINE (9)** printed pages (including cover page and appendix).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, while the assignment is in session, briefly explain your understanding of and assumptions about that question before attempting it.

4. You are to include the following particulars in your submission: Course Code, Full Name and Student PI and name your submission file as - CourseCode_FullName_StudentPI.

5. For answers which are hand-written, ensure that the question number is clearly stated on each page. All uploaded hand written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

6. Please submit only ONE (1) file (<500 MB) in either **PDF/JPEG/WORD format** within the time-limit via Canvas [similar to Tutor Marked Assignment (TMA) submission]. If you do not submit within the time-limit, you would be deemed to have withdrawn (W) from the course. Appeal is **NOT** allowed.

7. To prevent plagiarism and collusion, your submission will be reviewed thoroughly by Turnitin, The Turnitin report will only be made available to the markers. The university takes a tough stance against plagiarism and collusion. Serious cases will normally result in the student being referred to SUSS's Student Disciplinary Group. For other cases, significant marking penalties or expulsion from the course will be imposed.

Answer all questions. (Total 100 marks)

Question 1

In order to effectively manage the advertising expense and make revenue projections, the manager of a company which sells children clothes online, collected data from 35 sales periods. The revenue (in million SGD) for the company (y) and advertisement costs (in million SGD) (x) are recorded for each sales period. You may assume data from all months are independent.

Using this data, the following regression model is estimated:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \ , \ \varepsilon_i \sim N(0, \sigma^2).$$

Here are some of the reported summary statistics:

$$\sum_i x_i y_i = 637.2153$$
$$\sum_i x_i^2 = 173.0992$$
$$\sum_i y_i^2 = 2743.15$$
$$\bar{x} = 2.115285$$
$$\bar{y} = 7.991667$$

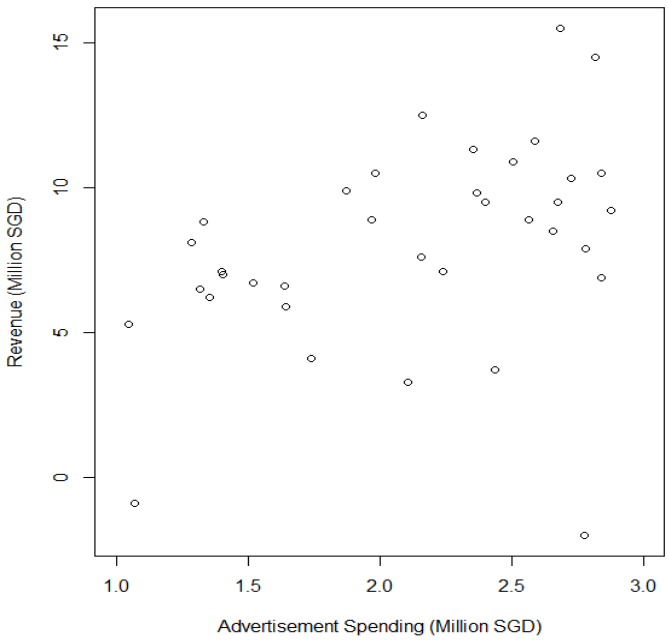


Figure Q1

(a) Estimate the coefficients β_0 and β_1 and interpret the point estimates.

(5 marks)

(b) Given that the estimated standard errors of β_0 and β_1 are 2.1024 and 0.9588 respectively, test the significance of the coefficients at $\alpha = 0.05$. Show all your steps.

(8 marks)

(c) State the distributional assumptions, if any, for the analysis in Question 1(b).

(2 marks)

(d) Suppose that there is a difference of 1.5 million SGD spent for advertisement in two randomly selected months.

(i) What is the expected difference in the revenue?

(2 marks)

(ii) Construct a 95 % CI for the difference in their expected response.

(8 marks)

Question 2

Accidents in aeroplanes are very rare but can lead to many fatalities in each incident. An analyst wishes to apply linear regression to analyze how the number of aeroplace incidents have changed from the period of 1988 to 1999 to the period of 2000 to 2014.

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \text{ , } \varepsilon_i \sim N(0, \sigma^2)$$

Where $y_i = incidence\ rate\ for\ airline\ i\ in\ 2000\ to\ 2014$
$x_i = incidence\ rate\ for\ airline\ i\ in\ 1988\ to\ 1999$

The sample size is 56, presenting 56 different airlines spanning across the globe. Also note that we are using incidence rate instead of number of incidents, because some airlines fly a lot more and thus should have higher number of incidents even with similar incidence rate.

The estimated equation is

$$\underset{(0.72297)}{E(y) = 2.88935} \quad + \quad \underset{(0.05324)}{0.15543\, x}$$

$$R^2 = 0.1363, \bar{x} = 7.915, \bar{y} = 4.119, \text{SSE} = 1043.755$$

However, the analyst feels that the simple linear regression model is not good enough. Thus, she proposes to include a new factor:

↗ k value = 10

$$E(y) = \beta_0 + \beta_1 x + \beta_2 (x - 10) z$$

↳ linear Spline

where

$$z = \begin{cases} 1, & \text{if } x > 10 \\ 0, & \text{else} \end{cases}.$$

With the addition of this new factor, the regression becomes a piecewise linear regression model. The estimates (standard error) for β_0, β_1 & β_2 are 1.2327(1.047), 0.538 (0.187) & -0.484(0.2275) respectively.

(a) Write down the equation of the models for cases $x > 10$ and $x \leq 10$.

(4 marks)

(b) Sketch E(y) against . x should lie on the horizontal axis. Label all key points of the plot.

(3 marks)

(c) Given that the SSE of this new model is 961.53, perform a nested F-test and determine if the new covariate is worth adding into the model.

(5 marks)

The analyst wishes to improve the model further by adding in one more covariate. The covariate "Asia" is an indicator variable tagged as 1 if the airline is headquartered in Asia, and 0 if not.

$$E(y) = \beta_0 + \beta_1 x + \beta_2 (x - 10)z + \beta_3 Asia$$

```
Residuals:
    Min      1Q  Median      3Q     Max 
-6.3030 -2.2161 -0.0762  1.4523 20.0225 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)   
(Intercept)   0.6565     1.0730   0.612  0.54331   
x             0.5016     0.1842   2.723  0.00879 **
(x-10)x1     -0.4366     0.2243  -1.946  0.05703 . 
Asian         2.1525     1.1852   1.816  0.07511 . 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.17 on 52 degrees of freedom
Multiple R-squared:  0.2518,	Adjusted R-squared:  0.2087 
F-statistic: 5.834 on 3 and 52 DF,  p-value: 0.001633
```

(d) Using the information from the R-output of the new model in Question 2(c), sketch this new model into the sketch drawn in part b. Label it clearly so that we can clearly see the differences between this model and the previous. Interpret the findings.

(8 marks)

(e) Instead of using a dichotomous variable "Asian", the analyst decides to have a new country variable. In the dataset, there are 56 airlines from 23 countries. Discuss whether this approach is feasible. If not, suggest a better way to include more country data into the model.

(5 marks)

$y_i - \beta_1(8 + x_i) = \varepsilon_i$

$[y_i - \beta_1(8 + x_i)]^2 = \varepsilon_i^2$

Question 3

In this question, we will handle some theoretical concepts in regression analysis.

(a) Consider this simple linear regression mode:

$$y_i = \beta_1(8 + x_i) + \varepsilon_i \text{ , } \varepsilon_i \sim N(0, \sigma^2), \sigma\{\varepsilon_i, \varepsilon_j\} = 0 \; for \; i \neq j.$$

Using calculus, prove that the ordinary least squares estimator for β_1 is

$$\hat{\beta}_1 = \frac{\sum_i y_i(8 + x_i)}{\sum_i (8 + x_i)^2}$$

(10 marks)

(b) In the big data era, analysts are handling increasingly large datasets – both in the sample size and the number of covariates. This question is about the variance of the predicted value, $\hat{y}$.

(i) Can the variance of the mean response be reduced to 0 if the sample size is large enough? (This is the variance used for finding confidence intervals)

(5 marks)

(ii) Can the variance of the predicted response be reduced to 0 with a large sample size? (This is the variance used for finding prediction intervals)

(3 marks)

(iii) Discuss the implications of Question 3(a) and 3(b).

(3 marks)

(c) Consider the multiple regression model:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \varepsilon_i.$$

Usually, when we use R to perform a regression analysis, we will use the lm-function to find all 4 betas.

In the situation where we already know that $\beta_2 = 20$, suggest how we can continue to use R to perform this analysis using standard code (such as the lm function).

(4 marks)

Question 4

An analyst is looking at a dataset which relates Wine scores with the amount of substances in the Wine. She wishes to perform diagnostics for leverage and influence on this data, and is asking you to help her with it.

y: Wine scores (from 1 to 10)
x_1: Citric Acid (sometimes added to wines to increase acidity)
x_2: Sulfites (antioxidant and antibacterial)
x_3: Alcoholic content

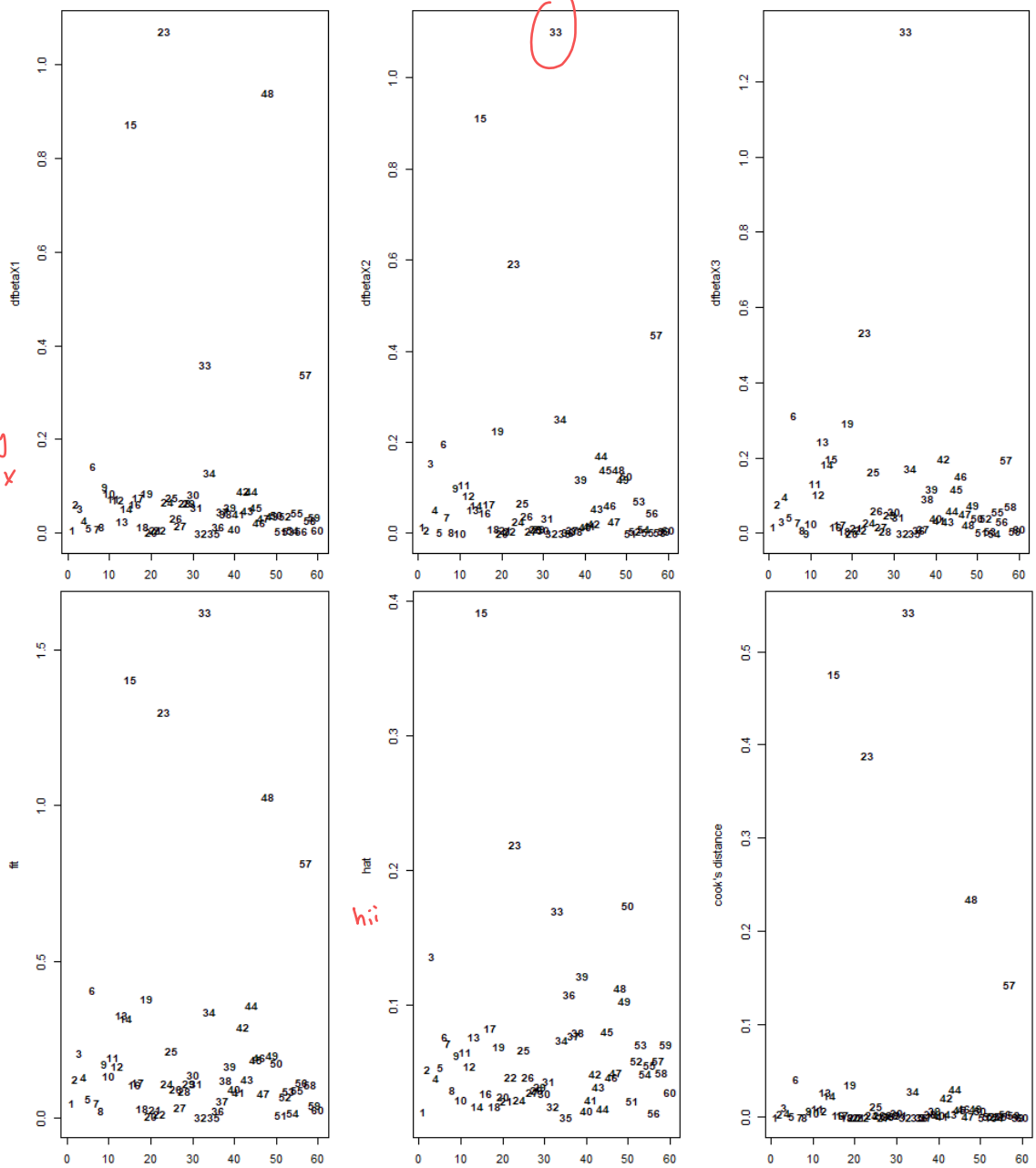
The regression model used is: $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \varepsilon_i$.

The slope coefficients are all positive and statistically significant, with $\hat{\beta}_2 > \hat{\beta}_3 > \hat{\beta}_1$.

The intercept coefficient is negative. $\rightarrow -\beta_0$

$\rightarrow$ data 33

$\rightarrow$ got 1 dfbeta for every single x



$\hat{y}$

h_{ii}

> 1

$2\bar{h} = 2\left(\frac{k+1}{n}\right)$

Figure Q4

The output from influence.measures ( ) in R is plotted and labelled with the sample number for your convenience. The top three are the DFBETAS for Citric Acid, Sulfites and Alcohol respectively. The bottom three are the DFFITS, diagonals of the Hat matrix and Cook's Distance respectively.

→ Make a table.

(a) Explain what is influence and what does an influential point do to your model?

(4 marks)

(b) Perform an analysis of influence on the individual coefficients, showing your working and justification clearly. Identify potential outliers.

(6 marks)

(c) Perform an analysis of influence on the model as a whole, showing your working and justification clearly. Identify potential outliers.

(6 marks)

(d) From your analysis in Question 4(b) and 4(c), you will have identified potential outliers. Also attached in this question is the raw dataset which was used by the analyst. By referring to the raw dataset, discuss on the outliers that you have observed.

(6 marks)

(e) What remedial measures do you recommend?

(3 marks)

in data, looks within range

but in the charts above, they are outliers.

Appendix A

n	Score	Acid	Sulf	Alco	n	Score	Acid	Sulf	Alco
1	4.5	0.1683	0.6328	12.7510	31	2.2	0.0264	0.4586	11.1107
2	2.2	0.0000	0.5932	10.7182	32	3.7	0.3507	0.7150	11.6732
3	7.3	0.0000	1.4164	14.2371	33	2.6	0.0000	0.3300	14.9833
4	3.6	0.3319	0.7379	10.7645	34	4.7	0.0000	0.3300	13.0373
5	2.4	0.1462	0.5011	10.2909	35	4.2	0.2184	0.7060	12.4846
6	6.2	0.0000	0.5683	14.1763	36	2.1	0.6744	0.4116	10.0751
7	3.9	0.1240	1.0400	11.2901	37	4.5	0.6990	0.6382	12.6159
8	3.5	0.4734	0.6862	11.5486	38	1.4	0.0525	0.4606	9.8261
9	3.4	0.5154	0.3300	11.5286	39	3.2	0.0000	1.0734	10.5814
10	4.3	0.0000	0.6605	11.9604	40	3	0.2582	0.6760	11.5741
11	5.2	0.0000	0.5125	13.7166	41	3.9	0.0495	0.7025	11.5950
12	5.1	0.0000	0.5463	13.5197	42	1.3	0.1035	0.5093	10.5170
13	7.4	0.2307	0.9863	14.8000	43	3.3	0.3872	0.4222	11.2374
14	3.1	0.2624	0.6977	13.1720	44	1.8	0.1423	0.5384	12.2803
15	7.2	0.9682	1.7351	14.8000	45	4.6	0.3748	0.3300	13.3413
16	4.8	0.0000	0.8545	12.3239	46	3.6	0.2935	0.6949	10.6030
17	3.6	0.6152	0.3300	12.0980	47	5.2	0.0134	0.6500	13.7419
18	4	0.0705	0.5799	12.4204	48	6.7	0.8321	0.7890	12.3442
19	5.9	0.0777	0.5023	13.9459	49	7.3	0.0721	1.2715	14.5382
20	3	0.0554	0.5144	11.4723	50	7.8	0.0000	1.5171	14.8000
21	3.5	0.1243	0.7088	11.3351	51	4.4	0.0305	0.6726	12.8877
22	3.3	0.0000	0.7437	11.1333	52	3	0.5124	0.5925	10.6401
23	2.5	0.9780	0.3300	13.2608	53	5.9	0.2403	1.2058	12.9878
24	4.4	0.0122	0.7465	11.9858	54	2.9	0.0000	0.3726	11.8342
25	6	0.0092	0.6802	14.2501	55	2.2	0.0000	0.5705	10.6047
26	3	0.3952	0.7722	10.9762	56	3.3	0.2287	0.5861	12.4923
27	5.1	0.3819	0.7281	13.3636	57	3.2	0.0000	1.0631	13.5835
28	4.4	0.5060	0.6421	12.2581	58	2.8	0.3629	0.5208	10.4389
29	2.5	0.0000	0.5677	11.2525	59	3.2	0.6648	0.5436	11.4785
30	2.7	0.0000	0.6478	11.4967	60	5.4	0.0948	0.8911	13.4889

outliers

----- END OF PAPER -----

Find CS of 1

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$, $\beta_2 = 3$

Let $\beta_0 + 3x_2$ be α_0

$y = \alpha_0 + \beta_1 x_1$

[reverse engineer in the last step]

$\downarrow$

equal $\begin{cases} \alpha_0 = _____ \\ \beta_0 + 3x_2 = _____ \end{cases}$